

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Winter Examination – 2022 Course: B. Tech. Branch :Electrical Engineering Semester : V Subject Code & Name: BTEEPLE504B Power Quality Issues Max Marks: 60 Date:08/02/2023 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Define power quality. Explain why power quality is important?		6
B)	Explain CBEMA and ITI Curve.		6
C)	Explain with neat diagram fundamental principle of protection of equipment against voltage sag.		6
Q.2	Solve Any Two of the following.		12
A)	Explain Power Quality evaluation procedure with flow Chart		6
B)	Explain with neat diagram how transients are produced by capacitor switching.		6
C)	State and explain any 6 IEEE Power Quality Standard for Harmonics		6
Q. 3	Solve Any Two of the following.		12
A)	Explain how Ferro resonance leads to Over voltage.		6
B)	Give the comparison of Harmonics and Transients		6
C)	What are the sources of Harmonics and explain its effects.		6
Q.4	Solve Any Two of the following.		12
A)	Explain the concept Triplen harmonics and explain its effect.		6
B)	Explain how the Lightning Phenomena leads to Over voltage Transients		6
C)	State the devices used for mitigation of overvoltage and explain any 2 with diagram.		6
Q. 5	Solve Any Two of the following.		12
A)	State and explain Objectives of power Quality Monitoring.		6
B)	State and explain selection criteria for power Quality Monitoring		6

	devices and state the advantages of Power Quality monitoring.		
C)	Name the equipment used for Power Quality Monitoring and explain any 2 Equipments.		6
	*** End ***		

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